

# IWERC'S Media Mention Tool

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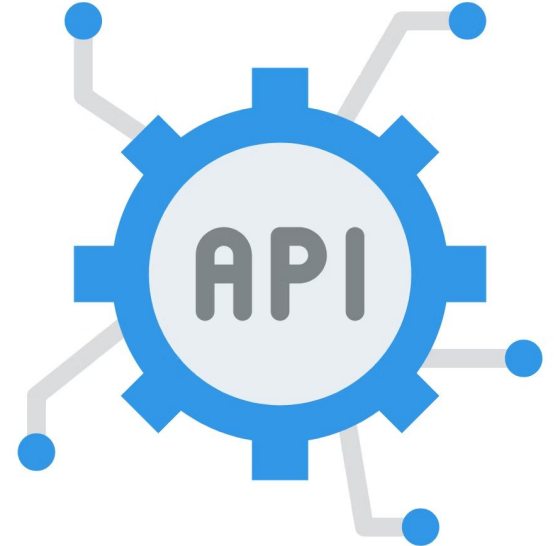
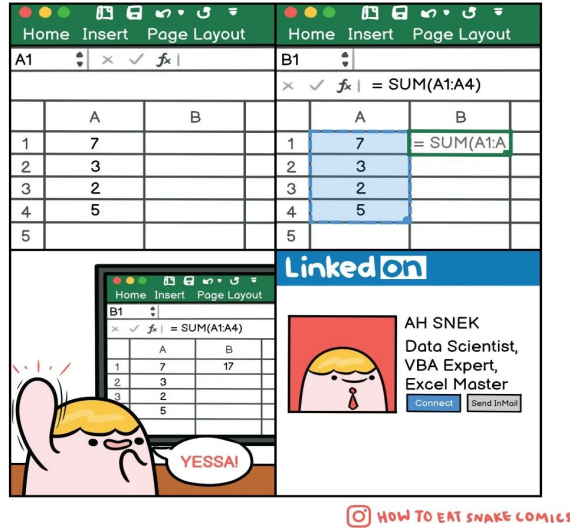
Research and Development /

## **Illinois Workforce and Education Research Collaborative (IWERC)**

*IWERC provides data-based analysis to our state's policy and education  
leaders' questions.*

# Issue Summary

# Objectives



## Overview:

IWERC (a division of DPI) focuses on advancing research, policy, and educational initiatives

## Challenges:

Manually tracking mentioned data using spreadsheets

Positive impacts are not visually represented on their website

## Solutions:

Dashboards

Web form

Automated Data Extraction Tool

# Recommendations

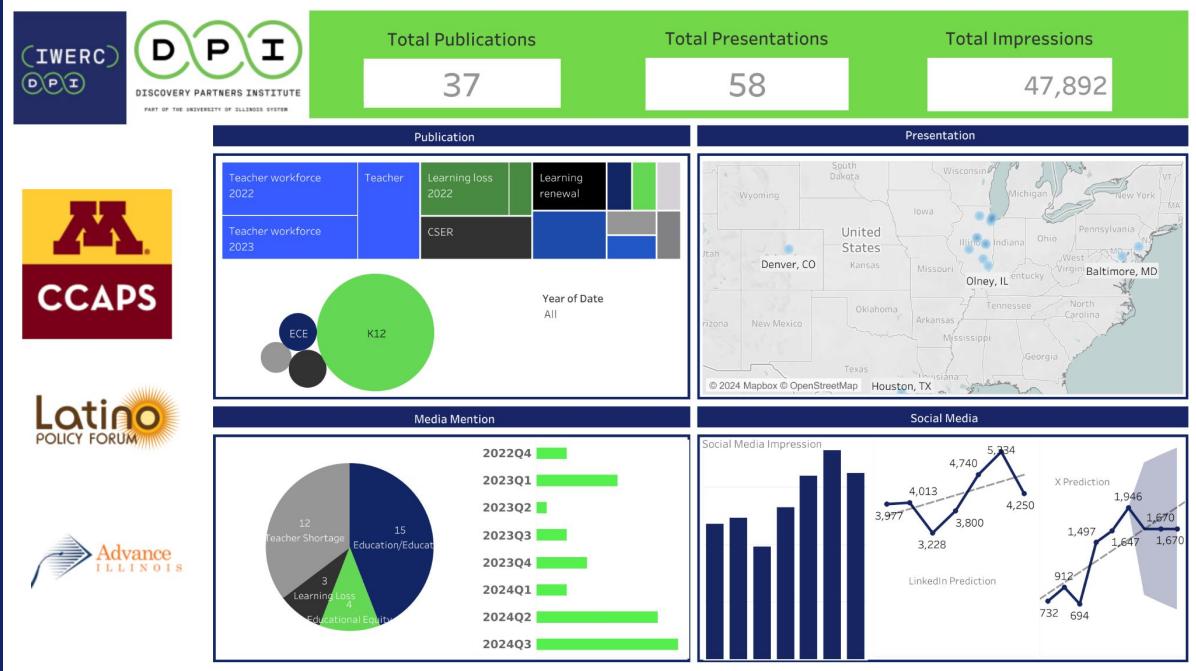
# Internal Tracking Dashboard

provides insights into IWERC's operations, highlighting yearly trends, publication/ presentation types, and monthly activity to optimize strategic initiatives.



# Website Dashboard

focuses on digital outreach, tracking social media impressions, geographic engagement, and keyword trends to enhance IWERC's impact and support its mission of advancing education and equity.



# Manual Web Form

## Manual Input Form

This web form allows researchers to input details about their publications or presentations. The entered data will automatically integrate with the Reporting Dashboard and Website Dashboard.

[Next](#)[Clear form](#)

- Introduce a manual entry form for researchers to input publication & presentation details
- Ensure accurate data capture and correct formatting for seamless integration into interactive dashboards

# Manual Web Form

Section 2 of 3

Publication

Description (optional)

Date of the Publication \*

Month, day, year

Citation

Long answer text

Publication

Section 3 of 3

Presentation

Description (optional)

Type of Presentation \*

☐ Public Presentation  
☐ Government Presentation

Title/Citation

Presentation



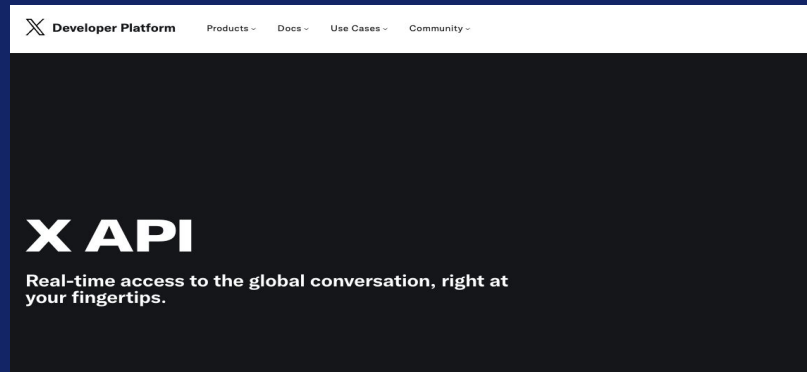
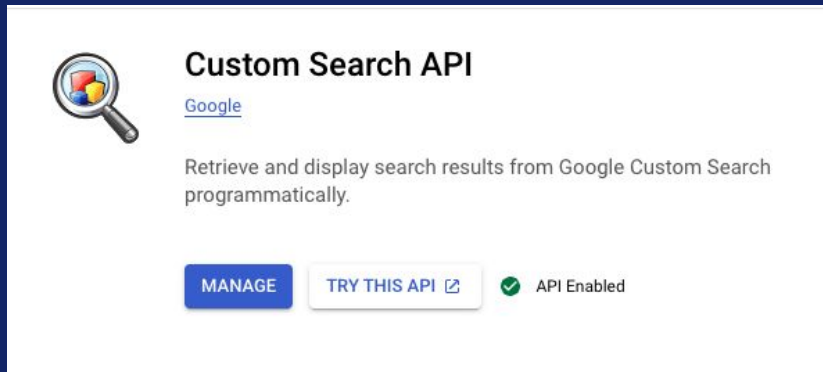
- All data collected from the form will be integrated into Google Sheet
- Data will be structured to make sure its integration

The screenshot shows a Google Sheet interface. The title bar reads 'Publication & Presentation'. The menu bar includes 'File', 'Edit', 'View', 'Insert', 'Format', 'Data', 'Tools', 'Extensions', and 'Help'. The toolbar shows search, undo, redo, print, and zoom (100%) options. The active cell is A33. The sheet contains a table with the following structure:

|    | A               | B             | C               |   |
|----|-----------------|---------------|-----------------|---|
|    | Form_Responses1 |               |                 |   |
| 1  | Timestamp       | Email Address | Input Data Type | D |
| 2  |                 |               |                 |   |
| 3  |                 |               |                 |   |
| 4  |                 |               |                 |   |
| 5  |                 |               |                 |   |
| 6  |                 |               |                 |   |
| 7  |                 |               |                 |   |
| 8  |                 |               |                 |   |
| 9  |                 |               |                 |   |
| 10 |                 |               |                 |   |
| 11 |                 |               |                 |   |
| 12 |                 |               |                 |   |
| 13 |                 |               |                 |   |
| 14 |                 |               |                 |   |
| 15 |                 |               |                 |   |
| 16 |                 |               |                 |   |
| 17 |                 |               |                 |   |
| 18 |                 |               |                 |   |
| 19 |                 |               |                 |   |
| 20 |                 |               |                 |   |

# Data Extraction Tool

Create an automated tool to gather data using Google Search API and social media APIs and store it in an online data sheet



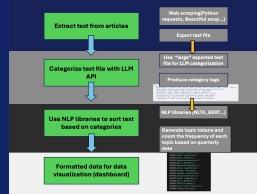
# How Data Extraction Tool Works?

## Step 1



Fetch social media  
& external website  
data through APIs

## Step 2



Sorting,  
Categorization  
and Data Cleaning

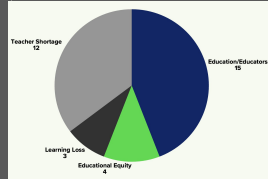
## Step 3



Store data in Google sheet

## Step 4

Import data and  
update graphs  
regularly



# Data Extraction (Step 2&3)

- Creating a programming script with **Python** that fetches search results from **Google Search API** to a **Google Sheet**
- Next step: data formatting to select and organize the right data to store in the spreadsheet



Social & Media Mention Extraction.ipynb

## Adding search results to Google spreadsheet

```
##import necessary files and libraries for google search result integration
from google.colab import auth
auth.authenticate_user()

from oauth2client.service_account import ServiceAccountCredentials
import gspread

# Load credentials for the service account
scope = ['https://spreadsheets.google.com/feeds', 'https://www.googleapis.com/auth/drive']
creds = ServiceAccountCredentials.from_json_keyfile_name('iwerconnect-c978736150b5.json', scope)
client = gspread.authorize(creds)

[] ## open google sheet file
sheet = client.open_by_url('https://docs.google.com/spreadsheets/d/13TxgoAY58C-kV6VxqCYu_huV1yfft9tEcZ_AbDP82Y/edit?usp=shari

[] ### using google search api
import requests
import re
import openpyxl
from datetime import datetime
search_api_key = 'AIzaSyB2eeu6R1lMrnQZGbmZDm72_FFYX0mniM40'
search_engine_id = 'f231d203aba714c16'
```

Result 1:

Date: 2024-12-10

Title: IWERC and the College of Education: Collaboration Discussion ...

Link: <https://education.illinois.edu/news/2023/06/07/iwer-and-the-college-of-e>

Snippet: Jun 7, 2023 ... All current faculty are invited to join the conversati

Result 2:

Date: 2024-12-10

Title: IL Workforce & Education Research Collaborative | Discovery ...

Link: <https://dpi.uillinois.edu/about/dpi-team/iwer/>

Snippet: IWERC's researchers and research associates collaborate with community

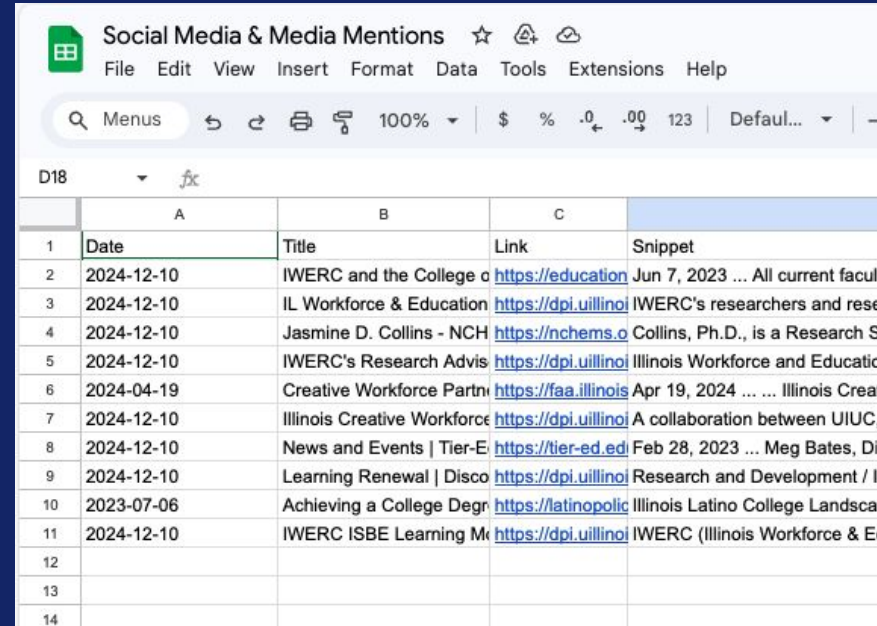
Result 3:

Date: 2024-12-10

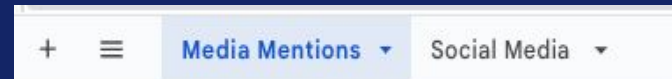
Title: Jasmine D. Collins - NCHEMS

# Data Extraction (Step 2&3)

- Both external data from websites and social media will be stored in this spreadsheet in our project account.
- The data must follow a specific format before being uploaded to the spreadsheet



| Social Media & Media Mentions                           |            |                             |                                                                                         |
|---------------------------------------------------------|------------|-----------------------------|-----------------------------------------------------------------------------------------|
| File Edit View Insert Format Data Tools Extensions Help |            |                             |                                                                                         |
| Menus                                                   |            |                             |                                                                                         |
| D18                                                     |            |                             |                                                                                         |
|                                                         | A          | B                           | C                                                                                       |
| 1                                                       | Date       | Title                       | Link                                                                                    |
| 2                                                       | 2024-12-10 | IWERC and the College o     | <a href="https://education">https://education</a> Jun 7, 2023 ... All current facul     |
| 3                                                       | 2024-12-10 | IL Workforce & Education    | <a href="https://dpi.uillinoi">https://dpi.uillinoi</a> IWERC's researchers and rese    |
| 4                                                       | 2024-12-10 | Jasmine D. Collins - NCH    | <a href="https://nchems.o">https://nchems.o</a> Collins, Ph.D., is a Research S         |
| 5                                                       | 2024-12-10 | IWERC's Research Advis      | <a href="https://dpi.uillinoi">https://dpi.uillinoi</a> Illinois Workforce and Educatio |
| 6                                                       | 2024-04-19 | Creative Workforce Partn    | <a href="https://faa.illinois">https://faa.illinois</a> Apr 19, 2024 ... Illinois Crea  |
| 7                                                       | 2024-12-10 | Illinois Creative Workforce | <a href="https://dpi.uillinoi">https://dpi.uillinoi</a> A collaboration between UIUC    |
| 8                                                       | 2024-12-10 | News and Events   Tier-E    | <a href="https://tier-ed.ed">https://tier-ed.ed</a> Feb 28, 2023 ... Meg Bates, Di      |
| 9                                                       | 2024-12-10 | Learning Renewal   Disco    | <a href="https://dpi.uillinoi">https://dpi.uillinoi</a> Research and Development / I    |
| 10                                                      | 2023-07-06 | Achieving a College Degr    | <a href="https://latinopolic">https://latinopolic</a> Illinois Latino College Landsc    |
| 11                                                      | 2024-12-10 | IWERC ISBE Learning M       | <a href="https://dpi.uillinoi">https://dpi.uillinoi</a> IWERC (Illinois Workforce & E   |
| 12                                                      |            |                             |                                                                                         |
| 13                                                      |            |                             |                                                                                         |
| 14                                                      |            |                             |                                                                                         |



# Media Text Categorization



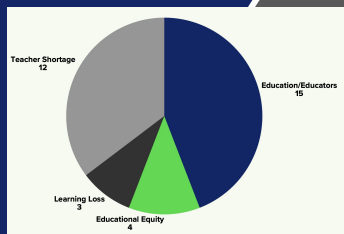
**Step 1**  
Find article sources

**Step 2**  
Extract raw text

Shereen Oca Beilstein and Meg Bates

reclines about teacher shortages abound as schools recover, conversations about teacher shortages are nothing new. These shortages begs the question: Are the factors that underpin this issue, we recently surveyed 3,478 current and former educators. A majority of educators feel their salary is low, less than half of current and former educators agreed that the biggest factor that motivates educators to leave is one-third of former educators felt supported by school leaders. COVID-19 safety was a concern for former educators. Current educators reported feeling safe from COVID-19. School policies must be transformed to align with the needs of educators of color, only one-fifth of those who were school leaders must cultivate supportive environments for early-career educators, less than half of those who were school leaders.

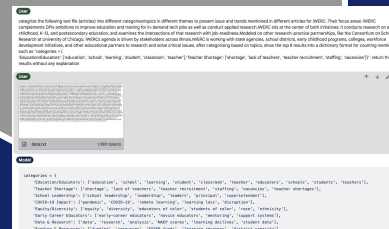
**Step 5**  
Data visualization (dashboard)



**Step 4**  
Sort raw text into different categories

```
2023Q1,Education/Educators,4
2023Q1,Teacher Shortage,4
2023Q2,Education/Educators,1
2023Q3,Learning Loss,1
2023Q3,Education/Educators,1
2023Q3,Educators of Color,1
2023Q4,Education/Educators,3
2023Q4,Educational Equity,2
2024Q1,Education/Educators,2
```

**Step 3**  
Generate categories with Gemini (LLM) API



# Extraction of Social Media Impressions

Create an app

\* Indicates required

App name\*

Enter your app name

LinkedIn Page\*

This action can't be undone once the app is saved.

Enter your company's name or LinkedIn Company Page URL (eg. <https://www.linkedin.com/company/...>)

For Third Party/Enterprise Developers: The LinkedIn Company Page you select will be associated with your app. Verification can be done by a Page Admin. Please note this cannot be a member profile page.

For Individual Developers: All products available to individual developers have a default Company page associated with them and you must select that default Company page to proceed.

To learn more about the products and the default Company pages, click here.

→ Create a new LinkedIn Page

Privacy policy URL

Begin with <http://> or <https://>

**Step 1**  
Register and get  
the APIs

**Step 2**  
Choose the best  
access of API version

## Find the right access for you

| Free                                                                                                                                                                                                                                                                                                                                             | Basic                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Pro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Enterprise                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| For write-only use cases and testing the X API                                                                                                                                                                                                                                                                                                   | For hobbyists or prototypes                                                                                                                                                                                                                                                                                                                                                                                                                                                   | For startups scaling their business                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | For businesses and scaled commercial projects                                                                                                                                                                                                                                                                                  |
| <ul style="list-style-type: none"><li>Rate-limited access to v2 posting and media upload endpoints</li><li>500 posts per month - posting limit at the user level</li><li>500 posts per month - posting limit at the app level</li><li>100 reads per month</li><li>1 app ID</li><li>Login with X</li><li>Access to Ads API</li><li>Free</li></ul> | <ul style="list-style-type: none"><li>Rate-limited access to suite of v2 endpoints</li><li>3,000 posts per month - posting limit at the user level</li><li>50,000 posts per month - posting limit at the app level</li><li>15,000 posts per month - read limit at the app level</li><li>2 app IDs</li><li>Login with X</li><li>Access to Ads API</li><li>\$200 per month</li><li>\$175 per month with annual subscription (\$2,100 total)</li><li>2 Top-ups allowed</li></ul> | <ul style="list-style-type: none"><li>Rate-limited access to suite of v2 endpoints, including search and filtered stream</li><li>288,000 posts per month - posting limit at the user level</li><li>300,000 posts per month - posting limit at the app level</li><li>1,000,000 posts per month - read limit at the app level</li><li>3 app IDs</li><li>Login with X</li><li>Access to Ads API</li><li>\$5,000 per month</li><li>\$4,500 per month with annual subscription (\$54,000 total)</li><li>2 Top-ups allowed</li></ul> | <ul style="list-style-type: none"><li>Commercial-level access that meets your and your customer's specific needs</li><li>Managed services by a dedicated account team</li><li>Complete streams: replay, engagement metrics, backfill, and more features</li><li>Access to Ads API</li><li>Monthly subscription tiers</li></ul> |
| <a href="#">Get started</a>                                                                                                                                                                                                                                                                                                                      | <a href="#">Subscribe now</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">Subscribe now</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <a href="#">Apply now</a>                                                                                                                                                                                                                                                                                                      |

**Step 3**  
Request the API  
and extract the  
data

```
# LinkedIn API URL
url = "https://api.linkedin.com/v2/organizationPageStatistics?q=organization&organs.

headers = {
    "Authorization": "Bearer YOUR_ACCESS_TOKEN"
}

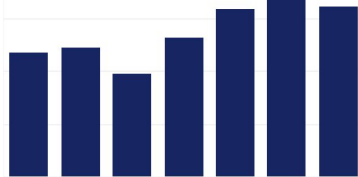
response = requests.get(url, headers=headers)
if response.status_code == 200:
    data = response.json()
    print(data) # 数据包含展示次数、点击率等
else:
    print("Error:", response.status_code)
```

**Step 4**  
Clean raw data  
and standardize  
date format

**Step 5**  
Data visualization  
(dashboard)

Total Impressions by Month

DATE



# API Plan Recommendation

 DEVELOPERS

Products

Docs and tools ▾

Resources ▾

My apps ▾

---

## Create an app

\* Indicates required

**App name \***

**LinkedIn Page \***

 This action can't be undone once the app is saved.

For Third Party/Enterprise Developers: The LinkedIn Company Page you select will be associated with your app. Verification can be done by a Page Admin. Please note this cannot be a member profile page.

For Individual Developers: API products available to individual developers have a default Company page associated with them and you must select that default Company page to proceed.

To learn more about the products and the default Company pages, click [here](#).

[+ Create a new LinkedIn Page](#) 

**Privacy policy URL**

**App logo \***

This is the logo displayed to users when they authorize with your app

## Free

For write-only use cases and testing the X API

- Rate-limited access to v2 posting and media upload endpoints
- 500 posts per month - posting limit at the user level
- 500 posts per month - posting limit at the app level
- 100 reads per month
- 1 app ID
- Login with X
- Access to Ads API
- Free

Get started

## Basic

For hobbyists or prototypes

- Rate-limited access to suite of v2 endpoints
- 3,000 posts per month - posting limit at the user level
- 50,000 posts per month - posting limit at the app level
- 15,000 posts per month - read limit at the app level
- 2 app IDs
- Login with X
- Access to Ads API
- \$200 per month
- \$175 per month with annual subscription (\$2,100 total)
- 2 Top-ups allowed

Subscribe now



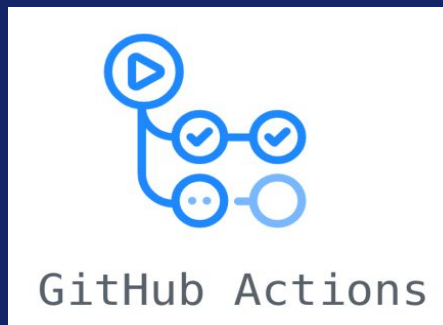
# Automation Recommendation

**Issue:** The code currently requires manual execution

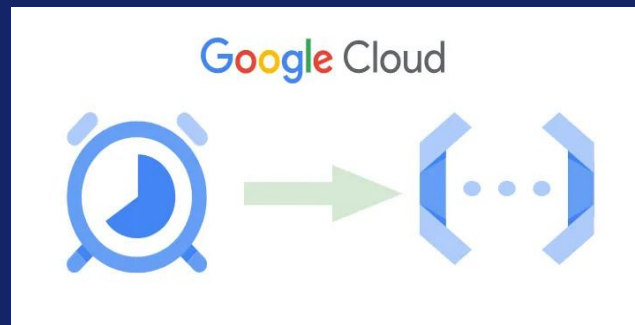
**Next step:** Implement automated (scheduled) execution solutions



1. Free tier includes 1 million requests/month.
2. Fully serverless and scalable.

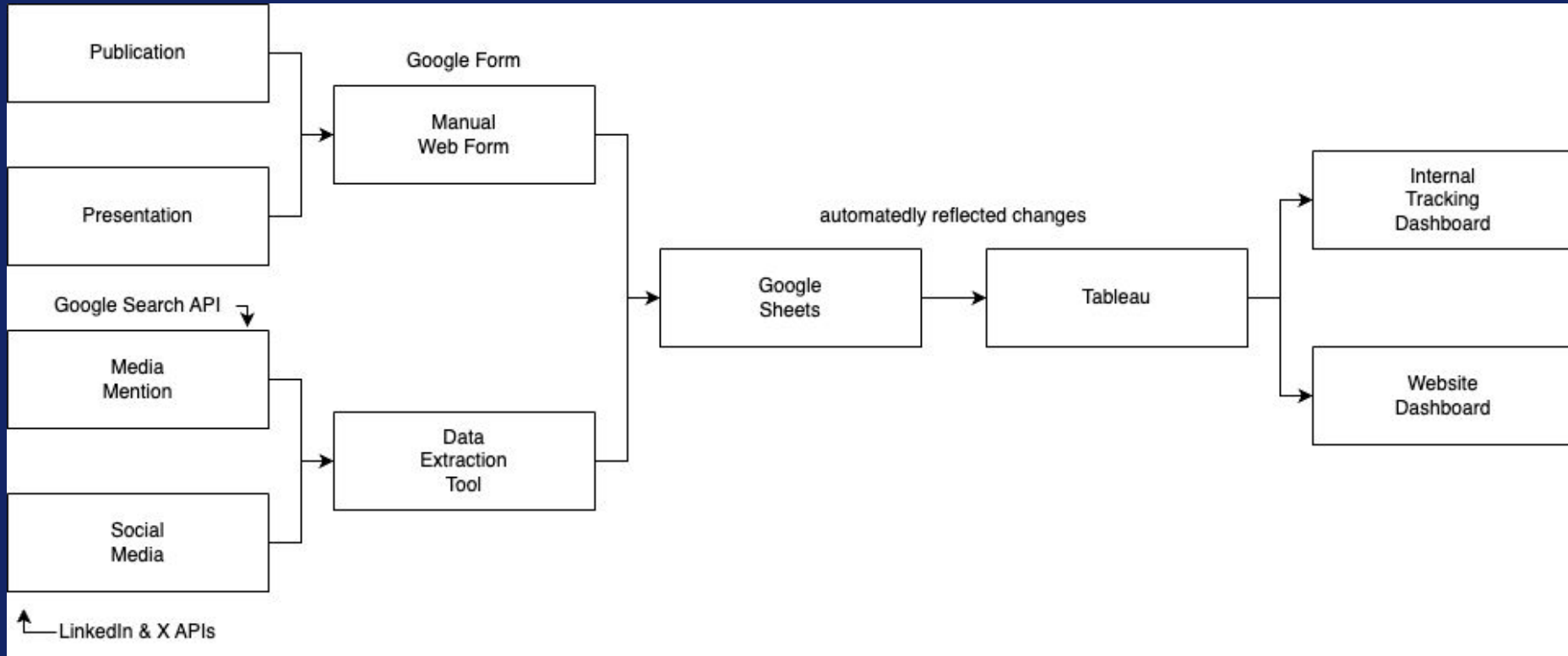


1. Provides 2,000 minutes/month for private repositories on the free plan.
2. Allows team collaboration

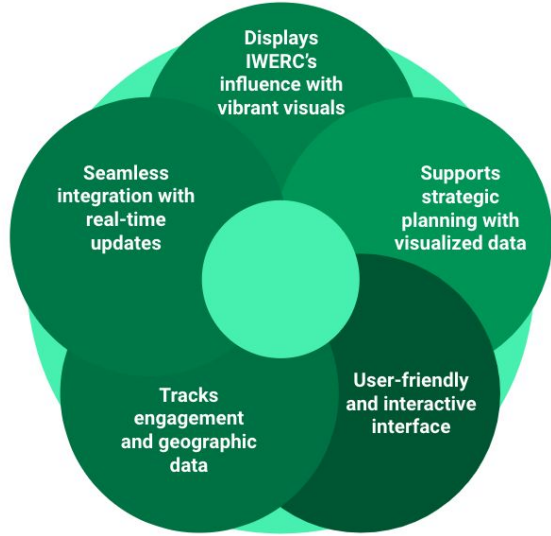


1. Free tier includes 2 million invocations/month for Cloud Functions.
2. Good for teams using Google Workspace.

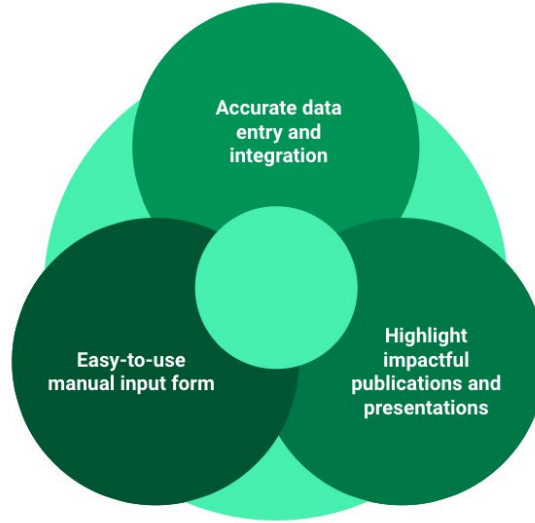
# How our Solutions Work?



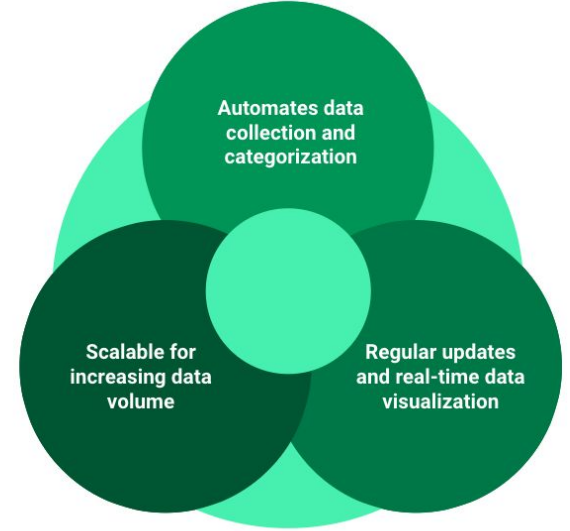
# Success Criteria



**Internal & Website  
Dashboards**



**Web Form**



**Data Extraction Tool**



# Thank You

@BADM 550

Do you have any  
questions?